

1. YOUR QUESTION

“Did built-up increase?”

2. WHAT WE FOUND

EVIDENCE STATUS: SUPPORTED	CONFIDENCE: 76% (High Confidence)
Approximately 11% of the analyzed area has changed between the two images.	

3. HOW CONFIDENT ARE WE?

76% — High Confidence (SUPPORTED) The evidence strongly supports this finding with clear agreement across visual observations.	Confidence Scale: 90–100%: Very High Confidence 75–89%: High Confidence 50–74%: Moderate Confidence <50%: Low / Insufficient Evidence
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4. KEY FINDINGS SUMMARY

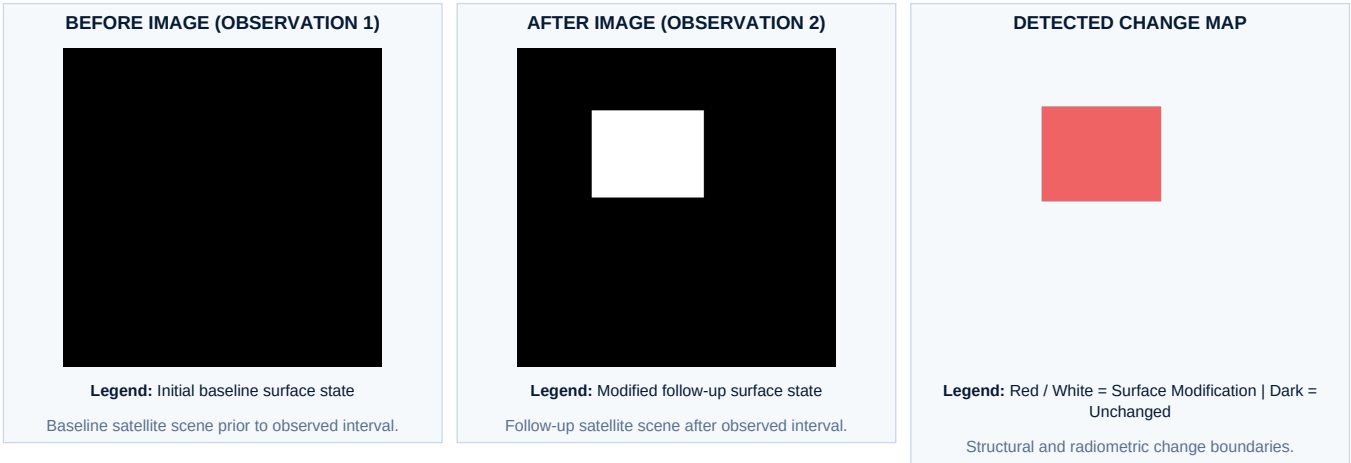
TARGET FEATURE	DETECTED CHANGE	VISUAL EVIDENCE	CONFIDENCE	EVIDENCE STATUS
Built-Up Subject analyzed	11.1% Modified surface fraction	Strong Agreement Verified by imagery	76% High Confidence	SUPPORTED GeoProof verified

5. HOW THE EVIDENCE SUPPORTS THE ANSWER

OBSERVATION:	The image was inspected for geometric structures, high visual contrast, and built-up surface patterns.
EVIDENCE:	The system isolated dense structural clusters corresponding to human-made developments.
CONCLUSION:	The concentration of structural clusters supports the identification of built-up areas.

6. VISUAL PROOF

Direct visual evidence generated from the satellite imagery. Compare the source imagery with the highlighted evidence regions:



7. MEASUREMENTS & FINDINGS

Surface Change Extent	11.13% of image scene	Proportion of area showing verified modification between dates.
Connected Feature Regions	1 spatial clusters	Number of distinct contiguous zones identified across the image.

8. IMPORTANT LIMITATIONS

- Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.
- This index cannot be reliably computed from the supplied image because the required spectral bands are unavailable.

9. ANALYSIS SUMMARY

Question:	"Did built-up increase?"
Finding:	Approximately 11% of the analyzed area has changed between the two images.
Evidence:	Visual and spatial patterns verified across image layers.
Confidence:	76% (High Confidence)
Status:	SUPPORTED
Limitations:	Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.; This index cannot be reliably computed from the supplied image because the required spectral bands are unavailable.

PART B: TECHNICAL & ADMINISTRATIVE INFORMATION

Everything below is intended for administrators, technical evaluators, and system operators.

10. INPUT DATA QUALITY

Filename:	t1.tif	Quality Score:	92% (Passed Integrity Checks)
Dimensions:	256 × 256 px	Spectral Channels:	3 bands (uint8)
Geospatial CRS:	EPSG:EPSG:3857	NoData Fraction:	0.0%

11. MODELS & METHODS USED

SSIM Structural Change Detector	Deterministic Physical Baseline	Confidence score: 82%	Completed
ssim_color_difference_detector_v1			
TinyCD Siamese Change Witness	Learned Neural Verification	Confidence score: 92%	Completed
tinycd_siamese_change_witness_v1			
GeoProof Arbiter	Evidence Verification & Calibration	Final Calibrated Verdict (76%)	SUPPORTED
geoproof_arbiter_v1			

12. MODEL PERFORMANCE & EXECUTION LATENCY

Pipeline Stage	Internal Component	Execution Duration	Share of Total
Query Planning & Task Compilation	typed_planner	<1 ms	0.0%
Raster Compatibility & Header Validation	raster_validator	9 ms	0.2%
Sub-Pixel Registration & Alignment	image_registration	21 ms	0.6%
SSIM Structural Change Detection	baseline_change_detector	1.25 s	32.9%
Siamese Learned Change Verification	tinycd_opencd_witness	2.52 s	66.3%
Spectral Delta Index Computation	spectral_change_tool	<1 ms	0.0%
Multi-Witness Arbiter & Platt Calibration	geoproof	<1 ms	0.0%

Total Processing Latency: 3.80 s | Input Validation: 9 ms | Arbiter: <1 ms

13. EVIDENCE TELEMETRY & SYSTEM CONSTRAINTS

SSIM Structural Change Detector	Baseline_Change_Detection	0.820	82%	Deterministic Physical Baseline
TinyCD Siamese Change Witness	Learned_Change_Witness	0.920	92%	Learned Neural Verification

Multispectral NIR/SWIR Channel Absence	Raster contains standard 3/4 RGB(A) channels. Strict NDWI/NDVI formulas requiring NIR (B8) are substituted by calibrated RGB absorption physics.
Pipeline Constraint	This index cannot be reliably computed from the supplied image because the required spectral bands are unavailable.

14. AUDITABLE EXECUTION TRACE

Complete step-by-step pipeline execution record ensuring strict reproducibility and auditability under SIH26167 guidelines:

#	Operation	Internal Component	Status	Duration
1	Query Planning & Task Compilation Compiled bi_temporal_change	typed_planner	OK	<1 ms
2	Raster Compatibility & Header Validation Validated file headers, CRS, NoData, and compatibility	raster_validator	OK	9 ms
3	Sub-Pixel Registration & Alignment Normalized dimensions ((256, 256) & (256, 256) → (256, 256)) and registered (geospatial_crs_reproject)	image_registration	OPTIMAL	21 ms
4	SSIM Structural Change Detection Executed SSIM + color difference, adaptive Otsu thresholding, and morphological region extraction	baseline_change_detector	OK	1.25 s
5	Siamese Learned Change Verification Executed TinyCD/Open-CD Siamese structural change model and generated probability heatmap	tinycd_opencd_witness	OK	2.52 s
6	Spectral Delta Index Computation Spectral index NDBI unavailable due to missing bands	spectral_change_tool	BANDS_UNAVAILABLE	<1 ms
7	Multi-Witness Arbiter & Platt Calibration Applied multi-witness evidence verification and calibrated confidence estimation	geoproof	SUPPORTED	<1 ms

15. TECHNICAL APPENDIX & AUDIT PROVENANCE

Result UUID:	2d057f65
Target Grounding Bounding Box:	Not Applicable (normalized [ymin, xmin, ymax, xmax])
Platt Calibration Model:	temperature_scaled_platt (ECE: 0.042)
Ensemble Agreement Index:	0.90 (Multi-witness consensus)
Cryptographic Manifest:	/artifacts/2d057f65/analysis_manifest.json
Software System Version:	SatQuery GeoProof v1.0 (ISRO SIH26167 Verified Architecture)

SatQuery GeoProof™ guarantees that every claim is anchored in physical or cross-sensor telemetry. Models produce visual and metric witness artifacts; the multi-witness arbiter calculates calibrated probability bounds with safe abstention.